

Hints for problems

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Only read this hints if you are stuck and want
some extra inspiration

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Problem 1: Entropy

Problem 1A (Discrete case)

The Shannon entropy can be written as

$$H(p) = -\left(s \log s + p \log p + (1 - s - p) \log(1 - s - p)\right).$$

To find its extrema, compute $H'(p) = 0$ and $H''(p)$.

Problem 1B (Continuous bounded case)

Hint 1

Try using the method of Lagrange multipliers. For a tutorial look at the file `lagrange_multiplier.pdf`

Hint 2

[Click to show/hide hint 2](#)